

María Teresa Valdivia-Mena

POSTDOCTORAL FELLOW, EUROPEAN SOUTHERN OBSERVATORY

Karl-Schwarzschild-Str. 2, 85748 Garching, Germany

☎ +49 89 32006289 | ✉ mariateresa.valdiviamena@eso.org, terevaldiviamena@gmail.com | 🏠 January 28th, 1995 in Santiago, Chile

Education

PhD (Dr. rer. nat) in Astrophysics

September 2020 - April 2024

MAX PLANCK INSTITUTE FOR EXTRATERRESTRIAL PHYSICS AND LUDWIG-MAXIMILIANS-UNIVERSITÄT

Garching, Germany

- Graduated with summa cum laude (final grade 0.7, German scale)
- Thesis title: *Asymmetric infall beyond natal cores to protoplanetary disks: Observations and analysis of streamers toward embedded low-mass protostars*

M.Sc. in Astronomy

March 2018 - August 2020

UNIVERSIDAD DE CHILE

Santiago, Chile

- Graduated with maximum distinction (Grade 6.8 out of a maximum of 7.0)
- Thesis title: *Molecular Clouds in Extreme Environments of the Low-Metallicity Magellanic System*

B.Sc. in Astronomy

March 2014 - December 2017

UNIVERSIDAD DE CHILE

Santiago, Chile

- Graduated with maximum distinction (Grade 6.2 out of a maximum of 7.0).
- Obtained a Minor in Scientific Computing together with the B.Sc.

Research experience

Postdoctoral Fellow

September 2024 - Present

EUROPEAN SOUTHERN OBSERVATORY

Garching, Germany

- Study the effects of asymmetric infall in the development of protostellar and protoplanetary disks.
- Study the relation between streamers' mass delivery and protostellar growth.
- Functional duties at the European ALMA Regional Center (ARC) at ESO Headquarters

Postdoctoral Researcher

May 2024 - August 2024

MAX PLANCK INSTITUTE FOR EXTRATERRESTRIAL PHYSICS AND LUDWIG-MAXIMILIANS-UNIVERSITÄT

Garching, Germany

- Part of the data calibration and analysis for the NOEMA large program PRODIGE (PI: P. Caselli and Th. Henning)
- Analysis of the kinematics of protostellar envelopes using interferometers (e.g. ALMA and NOEMA) and single-dish telescopes (e.g. IRAM 30m)
- Short bridging contract between end of PhD and start of Fellowship

Publications

As first author:

- Valdivia-Mena, M. T., Pineda, J. E., Caselli, P., et al., *Probing the physics of star formation (ProPStar). II. The first systematic search for streamers toward protostars*, 2024, *Astronomy & Astrophysics*, 687, A71
- Valdivia-Mena, M. T., Pineda, J. E., Segura-Cox, D. M., et al., *Flow of gas detected from beyond the filaments to protostellar scales in Barnard 5*, 2023, *Astronomy & Astrophysics*, 677, A92
- Valdivia-Mena, M. T., Pineda, J. E., Segura-Cox, D. M., et al., *PRODIGE - Envelope to disk with NOEMA I. A 3000 au streamer feeding a Class I protostar*, 2022, *Astronomy & Astrophysics*, 667, A12
- Valdivia-Mena, M. T., Rubio, M., Bolatto, A. D., et al., *ALMA resolves molecular clouds in metal-poor Magellanic Bridge A*, 2021, *Astronomy & Astrophysics*, 641, A97

As co-author (selection):

- Hsieh, T.-H. et al (incl. Valdivia-Mena, M. T.), *Resolved Gas Temperatures and 12C/13C ratios in SVS13A from ALMA Observations of CH3CN and CH3-13-CN*, 2025, accepted to *Astronomy & Astrophysics*
- Gieser, C. et al (incl. Valdivia-Mena, M. T.), *PRODIGE - Envelope to disk with NOEMA VI. An infalling gas bridge surrounding two Class 0/I systems in L1448N*, 2024, *Astronomy & Astrophysics*, 692, A55
- Pineda, J. E. et al (incl. Valdivia-Mena, M. T.), *Probing the physics of star formation (ProPStar). III. No evidence of dissipation of turbulence down to 20 mpc (4000 au) scale*, 2024, *Astronomy & Astrophysics*, 690, L5
- Pineda, J. E. et al (incl. Valdivia-Mena, M. T.), *Probing the physics of star formation (ProPStar). I. First resolved maps of the electron fraction and cosmic-ray ionization rate in NGC 1333*, 2024, *Astronomy & Astrophysics*, 686, A162
- Hsieh, T. H. et al (incl. Valdivia-Mena, M. T.), *PRODIGE - envelope to disk with NOEMA. III. The origin of complex organic molecule emission in SVS13A*, 2024, *Astronomy & Astrophysics*, 686, A289
- Semenov, D. et al (incl. Valdivia-Mena, M. T.), *PRODIGE - planet-forming disks in Taurus with NOEMA. I. Overview and first results for 12CO, 13CO, and C18O*, 2024, *Astronomy & Astrophysics*, 685, A126
- Hsieh, T. H. et al (incl. Valdivia-Mena, M. T.), *PRODIGE - Envelope to disk with NOEMA II. Small-scale temperature structure and streamer feeding the SVS13A protobinary based on CH3CN and DCN*, 2023, *Astronomy & Astrophysics*, 669, A137

Telescope proposals and experience

Atacama Large Millimeter/Submillimeter Array (ALMA)

Northern Extended Millimetre Array (NOEMA)

IRAM 30-m

Green Bank Telescope (GBT)

Atacama Pathfinder Explorer (APEX)

PI of accepted projects 2023.1.00572.S (grade B), 2022.1.00898.S (grade C)

PI of accepted project W22AH (grade A, 2022) and S25AI (grade B, 2025)

PI of accepted project 124-21 (2022A). On-site and remote observing experience.

PI, on-site and remote observer of accepted project GBT22B-163 (2022B)

PI of accepted project C104-9709 (2019B). On-site observer for Chilean time.

Teaching

Apr. 2025	Lecturer for “Introduction to Astronomy Data Analysis” , <i>Schulerpraktikum workshop (High school level)</i> , 2-day lesson on Python data analysis for ALMA datasets	<i>European Southern Observatory</i>
Sep. 2018 - Jul. 2019	Teaching Assistant for the course “Experimental Astronomy” , <i>Undergraduate course</i> , One semester, includes giving classes, marking and supervising activities	<i>Universidad de Chile</i>
Mar. 2019 - Jul. 2019	Teaching Assistant for the course “Introduction to Cosmology” , <i>Undergraduate course</i> , One semester, includes giving classes, marking and supervising activities	<i>Universidad de Chile</i>
Jan. 2019	Teaching Assistant for “Planetary Sciences” , <i>High School level</i> , one month course at “Escuela de Verano” (Summer school)	<i>Universidad de Chile</i>
Mar. 2017 - Jul. 2017	Teaching Assistance for the course “Mechanics” , <i>Undergraduate course</i> , One semester, includes giving classes, marking and supervising activities	<i>Universidad de Chile</i>

Outreach

	Member of “Cazadoras de Estrellas” Project , <i>DIRVEX, Universidad de Chile</i> . This outreach project is lead by a group of female graduate students and alumni from the Astronomy Department in Universidad de Chile.	
Jul. 2018 - Jan 2023	The goal is to give high school girls a hands-on experience with astronomical data and meet female astronomers so that they can solve any vocational doubts about this career	<i>Chile</i>
Aug. 30th 2022	Public Talk “Discos protoplanetarios, el origen de los mundos” , Programa Ciencia Abierta, Par Explora RM Norte. Liceo Bicentenario Provincial Santa Teresa de los Andes	<i>Colina, Chile</i>
Jul. 23rd 2022	Public Talk “How to feed baby stars” , <i>Soapbox Science Munich</i>	<i>Munich, Germany</i>
Jun. 27th, 2019	Public Talk “Eclipses: Luces y Sombras en el Universo” , <i>Municipalidad de El Bosque</i> .	<i>Santiago, Chile</i>
Mar. 21st 2019 - Apr. 9th 2019	Public Talk “Astroinformática para principiantes” , Done at DUOC UC Puente Alto and at Centro de Extensión de la Universidad de Talca.	<i>Santiago, Chile</i>
	Scientific Counselor for “Festival de Teatro y Ciencias” (FETYC) , <i>Par Explora Sur Oriente, CONICYT</i> . The FETYC is a theatre competition organized by the National Commission of Scientific and Technologic	
Jun. 2018 - Nov. 2018	Investigation (CONICYT) where several schools prepare a play around a scientific inquiry. The Scientific Counselor guides the scientific content of the plays.	<i>Santiago, Chile</i>
Mar. 2018 - Aug. 2018	Tourist Guide at National Astronomical Observatory , <i>Departamento de Astronomía, Universidad de Chile</i> .	<i>Santiago, Chile</i>

Scholarships and Awards

June 2025	Otto Hahn Medal , <i>Max Planck Society</i> . Award recognizing outstanding scientific achievements in connection with the doctorate work.	
September 2024	ESO Fellowship , <i>European Southern Observatory, Garching</i> . 3-year contract with functional duties.	
July 2018	Adelina Gutierrez Grant , <i>Sociedad Chilena de Astronomía</i> . The grant funded the assistance to the Second Binational AAA-SOCHIAS Meeting	
March 2018	Scholarship for National Master Studies , <i>Comisión Nacional de Investigación en Ciencia y Tecnología (CONICYT)</i> . The scholarship funds master studies done in Chilean universities	

Service and Leadership

May 2025 - Present	Fellow Representative , Office for Science, European Southern Observatory
Apr. 2019 - Apr. 2020	Graduate Student Representative , Departamento de Astronomía, Universidad de Chile

Talks

SEMINARS

Feb. 5th 2025	ORIGINS RU-D day: Filamentary structures at all scales , Title: “Rivers in the sky: The first systematic search for streamers toward protostars”, ORIGINS Excellence Cluster	Garching, Germany
Nov. 20th 2024	Cardiff Astro Seminar Series , Title: “Rivers in the sky: The first systematic search for streamers toward low-mass protostars”, School of Physics and Astronomy, Cardiff University	Cardiff, Wales, UK
Sep. 17th 2024	Star and Planet Formation Seminar , Title: “Are Streamers a Common Phenomenon? The first systematic search of streamers in a star-forming Region”, ESO	Garching, Germany
Nov. 29th 2023	Seminarios de Física , Title: “Rivers in the sky: Disrupting Symmetry in Star Formation Through Asymmetric Infall onto Protostars”, Departamento de Física, Universidad de Santiago de Chile	Santiago, Chile
Sep. 13th 2023	Departamento de Astronomía Seminar , Title: “Rivers in the sky: Disrupting Symmetry in Star Formation Through Asymmetric Infall onto Protostars”, Departamento de Astronomía, Universidad de Chile	Santiago, Chile
Dec. 12th 2022	Tuesday UVA / NRAO Astronomy (TUNA) Lunch Talk , Title: “Rivers in the sky: streamers discovered towards two young embedded sources in Perseus”, NRAO	Charlottesville, USA
Dec. 12th 2022	IRAM-Wisc-CAS Seminar , Title: “Rivers in the sky: streamers discovered towards two young embedded sources in Perseus”	Virtual

CONFERENCES

May 7th 2025	Puzzles of star formation , Title: “Are the physical processes shaping molecular clouds affecting disk formation and evolution?”	Ringberg Castle, Germany
June 28th 2024	The promises and challenges of the ALMA Wideband Sensitivity Upgrade , Title: “Rivers in the sky: streamers as catalysts for physical and chemical changes in protostars and disks”	ESO, Garching, Germany
May 15th 2024	Early Phases of Star Formation , Title: “The first systematic search for streamers toward low-mass protostars”	Ringberg Castle, Germany
Feb. 20th 2024	Celebrating 30 Years of Protoplanetary Disk Chemistry: past, present, and future , Title: “Streamers: catalysts for physical and chemical changes in protoplanetary disks”	Ringberg Castle, Germany
Aug. 4th 2022	From Clouds to Planets II: The Astrochemical Link , Title: “Rivers in the sky: streamers discovered towards two Class I sources in Perseus”	Berlin, Germany
Jun. 30th 2022	European Astronomical Society Annual Meeting 2022 , Title: “Rivers in the sky: Streamers discovered towards two embedded protostars in Perseus”	Valencia, Spain
Mar. 4th 2022	Meeting of ALMA Young Astronomers , Title: “Rivers in the sky: streamers discovered towards two Class I sources in Perseus”	Virtual
Nov. 5th 2021	12th IMPRS Students Symposium , Title: “River in the sky: A streamer feeding a Class I protostar”	Garching, Germany
Oct. 29th 2021	Gaps, Rings, Spirals, and Vortices: Structure Formation in Planet-Forming Disks , Title: “A Class I protostar with a high streamer mass infall rate”, Munich Institute for Astro- and Particle Physics	Garching, Germany
Jun. 30th 2021	European Astronomical Society Annual Meeting 2021 , Title: “River in the sky: A streamer feeding a Class I protostar”	Virtual
Nov. 8th 2019	Latin American Regional IAU Meeting 2019 , Title: “ALMA observations of molecular tracers in R136 in 30Dor: How does cold gas survive?”, Hotel Antofagasta	Antofagasta, Chile

Skills

Languages	Spanish (native), English (C1), German (A2)
Programming languages	Python (expert), C (beginner), R (beginner)
Astronomy tools	Data reduction and imaging with CASA and GILDAS (CLASS, CLIC, MAPPING) for interferometers and single-dish telescopes
Version controlling tools	Git, Google Docs

References

Prof. Dr. Paola Caselli, Scientific Director, Max Planck Institute for Extraterrestrial Physics,
caselli@mpe.mpg.de

Prof. Dr. Mónica Rubio, Full Professor, Departamento de Astronomía, Universidad de Chile,
mrubio@das.uchile.cl

Dr. Jaime E. Pineda, Staff Scientist, Max Planck Institute for Extraterrestrial Physics, *jpineda@mpe.mpg.de*